SectraTEK

AI FOR CYBERSECURITY AND DIGITAL FORENSICS · LAB 07

Ask the Device



A small agent reads an entire Android extraction for you: ask a plain-English question and get an answer grounded in the image, with the exact file it used.

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YOUR INSTRUCTOR

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IN ONE LINE

I study how phones get spied on and tampered with, and build AI tools that help examiners find the evidence.

Mobile threat research
Spyware, stalkerware and device tampering.

AI for investigators
Machine learning that speeds up the search.

Teaching
AI for Cybersecurity and Digital Forensics.

TODAY'S STORY

From a whole phone image to one cited answer

1

Background

One answer hides in thousands of files.

2

Before AI

Find the artifact, then hand-write SQL.

3

The problem

Expert-bound and slow, or the model guesses.

4

The gap

A chatbot knows Android, not your phone.

5

How agents help

Give the model tools, and let it loop.

6

The method

List, read, query: think, act, observe.

7

The case

Which accounts are on this Pixel? The agent finds it.

8

Your lab

Build the agent by hand in the browser.

Practical runs in Google Colab with a free hosted Qwen model. Nothing is installed on your computer.

BACKGROUND

The examiner's needle in a haystack

A full phone extraction is thousands of files and dozens of databases. The answer to a simple question sits in exactly one of them, and you have to know where.

A whole filesystem

App data, system files, media and logs: gigabytes per device.

Dozens of databases

Accounts, contacts, SMS, chat apps: each its own SQLite schema.

One artifact matters

“Which accounts are on this phone?” is answered by a single account record.

BEFORE AI

The old way: find the file, then write SQL

1. Identify the location

You have to know that the Google account lives in a specific database, deep in the app-data tree, among thousands of other paths.

```
data/data/com.google.android.gm/
    databases/accounts.db
```

2. Write the SQL

Then you open that database and write a query by hand to pull the record you want out of the right table.

```
SELECT name, type
FROM accounts;
```

THIS IS EXACTLY HOW IT HAS BEEN TAUGHT

“I used to teach students to identify the correct location and use SQL to retrieve the answer.” It works, and it is precise, but it needs an expert who already knows the map of the device.

THE PROBLEM

The shortcut, asking a model directly, invents the answer

ASK A MODEL: WHICH ACCOUNTS ARE ON THIS PHONE?

“Based on a typical Android device, most likely it is john.doe@gmail.com.”

NOTHING BEHIND IT

It never saw this device

The model knows Android in general, not the contents of your extraction.

So it guesses

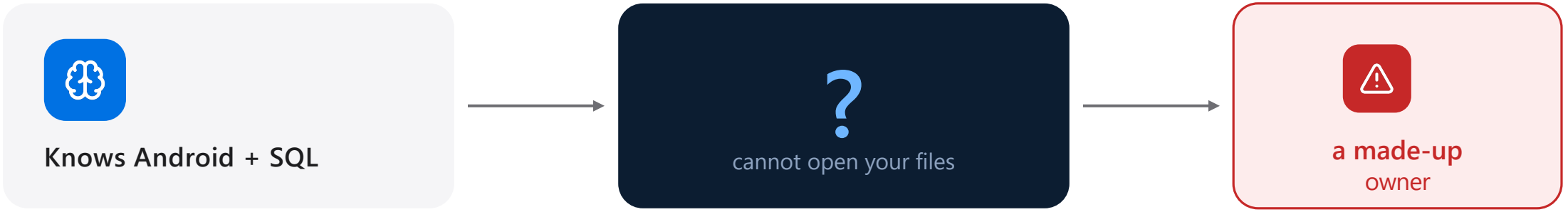
Asked which accounts are on the phone, it offers a fluent, plausible, fictional name.

No file to check

There is no artifact behind the answer, nothing you can verify or cite.

THE GAP

Knowledge, but no hands

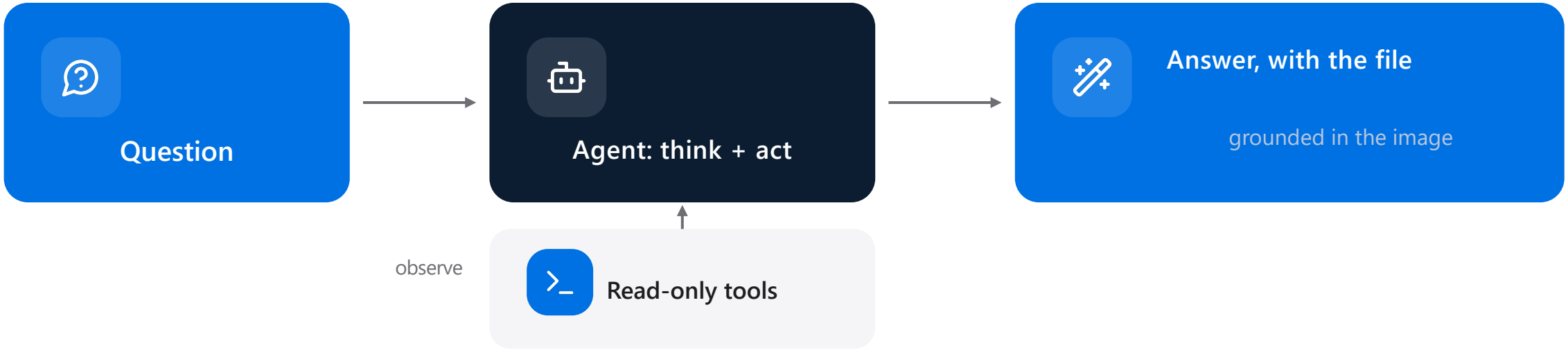


GIVE THE MODEL HANDS

The fix is not a bigger model. It is to give the model tools so it can open the evidence itself, look at what it finds, and answer only from that. A model that can act on the device, not just talk about it, is an agent.

THE ANSWER

Think, act, observe, repeat



IT ONLY SEES WHAT THE TOOLS RETURN

It answers from the files it actually opened, not from memory. That is what keeps it honest.

AND IT CITES THE FILE

Every answer names the exact artifact it came from, so you can open it and check.

HOW IT WORKS

Four moves, built by hand

Think

Read the question and the last result, and decide the next move.

Act

Call one read-only tool: list a folder, read a file, or run SQL.

Observe

Take the tool's output as a new fact, and reason from it.

Answer

When the evidence is enough, answer and cite the file.

YOU BUILD THE LOOP YOURSELF

No framework hides the agent. In the notebook the three tools and the think-act-observe loop are a few lines of plain Python, so you can see exactly how each decision is made and where each answer comes from.

THE GUARDRAIL

Three tools, and a hard boundary

The agent can do only three things, and all three are read-only and locked inside the image folder.

list_dir(path)	See what is inside a folder.
read_file(path)	Read the text of a file.
run_sql(path, query)	Run one read-only SELECT on a database.

WHY A BOUNDARY

Read-only, and sealed inside the image, so the agent cannot change evidence or wander off the device, the way a loosely-scoped agent drifts into fetching a web page it was never meant to touch.

The same discipline as handling the real exhibit: look, never alter, and stay within the evidence.

THE PRACTICAL

One question, the whole image

WHAT'S IN THE IMAGE

- Gmail / Google account
- Contacts
- SMS messages
- WhatsApp messages
- Installed apps
- Saved Wi-Fi networks
- Device properties

WHAT THE AGENT DOES

Walk the image

Lists folders on its own until it finds the account database.

Query the artifact

Writes the SQL to read the Google account record.

Answer with a path

Names the owner and the exact file the answer came from.

SEE THE DIFFERENCE

The same question, two ways

Model alone

no device

"Most likely john.doe@gmail.com."

Fluent, invented, and no file to check.

With the agent

read + cited

"The signed-in accounts are nadia.okafor@gmail.com and a WhatsApp number."

From the real account database, and it cites accounts.db.

PRACTICAL

Your turn, in the browser

1	**Open in Colab**	Click the badge in the notebook. Nothing to install.
2	**Paste a free key**	Get a free OpenRouter key for the Qwen model, paste when asked.
3	**Build the image**	Create the small synthetic Android extraction in one cell.
4	**Give it tools**	Define the three read-only tools and the think-act-observe loop.
5	**Ask and trace**	Ask questions, watch it navigate, and check the cited file.

TAKE IT FURTHER

Point Sectra at a real extraction

The lab image is synthetic so it runs instantly. Sectra walks whatever tree you hand it, so you can aim it at real evidence three ways:

Synthetic (default)	Built in one cell. Instant and identical for everyone - what the lab uses.
A real public image	A Pixel 3 / Android 13 full file system image from Digital Corpora (academic, no login).
Bring your own	Already have an extraction folder? Call use_extraction('/path') and ask away.

REAL FULL-FILE-SYSTEM IMAGES ARE BIG - THAT IS HONEST, NOT A LIMITATION

The public Pixel 3 image is about 18.5 GB to download and ~40 GB extracted: fine on a laptop or a roomy runtime. The fetch cell is left commented out, so nothing downloads until you choose - the synthetic image stays the default.

SectraTEK

Let the device answer for itself.

A forensic agent turns a plain-English question into a guided search of the evidence, and shows you the exact file behind every answer.

